

VoxShield

Indic-Language Voice-Deepfake Detection under Telephony Conditions
A multi-detector, telephony-hardened prototype for financial-fraud call screening

CURRENT STATUS — early-stage research prototype.

We treat the results below as evidence/hypotheses to stress-test, not as claims of production readiness. We are writing specifically to ask whether our evaluation methodology is rigorous enough — particularly around cross-generator generalisation and telephony/codec robustness.

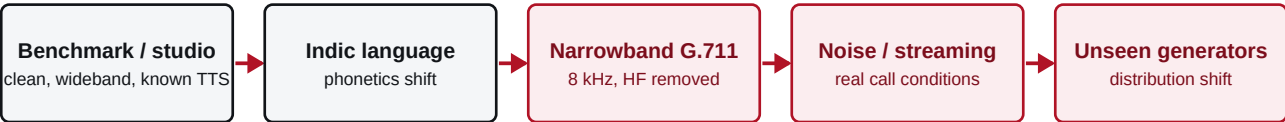
Strongest measured results (each with its evaluation scope).

Result	Value	Evaluation scope
Meta-fusion EER	5.9%	held-out In-the-Wild (cross-dataset)
Best single detector (XLS-R 300M)	16.0% EER	same evaluation set
Naive averaging	19.1% EER	same set (worse than best single)
Indic fake recall	42% → 82%	MMS-TTS-generated Indic fakes; English EER unchanged
Genuine-Indic false-positive rate	36.3% → 6.3%	tested Indic conditions; channel-aware
ROC-AUC · calibration	0.983 · ECE 0.044	held-out; calibration distribution

EER is not accuracy; Indic recall is not overall recall; the false-positive rate is not universal. Scopes are stated precisely and are not conflated.

1 The Research Question

Central question: can voice-deepfake detectors that perform well on benchmark / studio speech remain reliable when speech is Indic-language, narrowband (8 kHz G.711 μ -law), noisy, streamed, and produced by synthesis systems not seen during training? Each of these is a distribution shift, and they compound in exactly the setting we care about — a live fraud call to a bank.



Each step widens the gap between where detectors are trained and where fraud calls occur

Figure 1 — The domain-gap chain between benchmark conditions and real fraud calls.

Our interest is less 'is this clip synthetic?' in the abstract and more: under these compounded shifts, what evaluation protocol would let us (and a reviewer) trust a reported number?

2 System Architecture (what is implemented)

VoxShield is a streaming pipeline. Each block below is implemented in the current prototype.

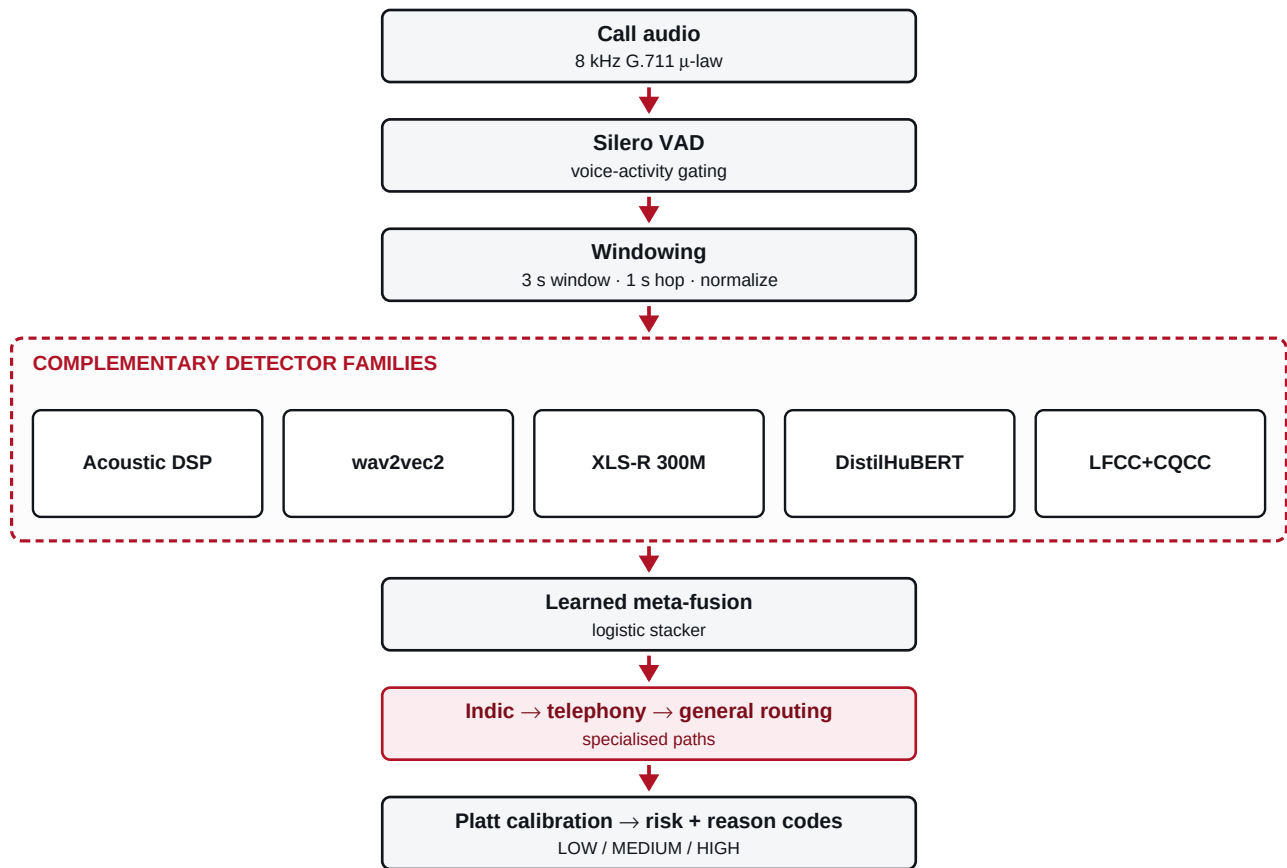


Figure 2 — Implemented inference pipeline: telephony audio → VAD → windowing → complementary detectors → learned fusion → routing → calibrated risk + reason codes.

Block	Input → process → output
Front-end	8 kHz G.711 μ-law → Silero VAD → 3 s windows / 1 s hop → normalized frames
Feature bank	frames → 89-dim vector at a 16 kHz analysis rate (LFCC, CQCC, group-delay phase, F0/prosody, jitter, shimmer, and a >6 kHz band)
Detectors	features/audio → 5 complementary detector scores (DSP, wav2vec2, XLS-R 300M, DistilHuBERT, LFCC+CQCC)
Fusion + routing	scores → learned logistic stacker → Indic/telephony/general routing → single score
Decision	score → Platt calibration → LOW/MEDIUM/HIGH + reason codes → fraud workflow

3 Feature & Model Design — Rationale

We combine detector families that fail in different ways. **This is a working hypothesis, not a proven independence result** — we have not yet run a formal detector-error correlation analysis.

- **A. Signal-level evidence** — LFCC, CQCC [12], group-delay phase, prosody (F0, jitter, shimmer), and a >6 kHz band. (Features are computed at a 16 kHz analysis rate, 8 kHz Nyquist; the >6 kHz band is informative only for wideband audio and is empty under 8 kHz G.711 — we treat that emptiness itself as a narrowband indicator rather than expecting HF artifacts there.)
- **B. Learned representations** — wav2vec2 [5], XLS-R 300M [6] (multilingual), DistilHuBERT [11]: self-supervised features with different pretraining.
- **C. Fusion** — a learned logistic meta-stacker over detector outputs (some weights negative), rather than naive averaging.
- **D. Routing** — Indic-aware → telephony/narrowband → general, selected by on-device language-ID.

The measured detector/fusion comparison (§5) is our first evidence for the fusion hypothesis; a per-family ablation under disjoint conditions is a validation target, not yet complete.

4 Telephony Hardening — tested vs engineered

Telephone audio discards most detail above ~3.4 kHz [4] — often exactly where synthesis artifacts live — and adds codec distortion, noise and packet loss. We build for this, but we are careful to separate what is engineered from what is measured:

Telephony condition	Status in this work
G.711 μ -law 8 kHz inference pipeline	Implemented
Narrowband / codec training augmentation	Implemented
Simulated narrowband (augmented) evaluation	Evaluated under augmented conditions
Real PSTN / SIP captured-call evaluation	Not yet performed (pending)

We would especially value your view on whether augmentation-based telephony evaluation is an acceptable proxy, and what a minimal real-network evaluation should look like.

5 Indic Experiment

We synthesised an Indic fake set with MMS-TTS [9], added an Indic-aware path with language-ID routing, and retrained. On the tested Indic conditions:

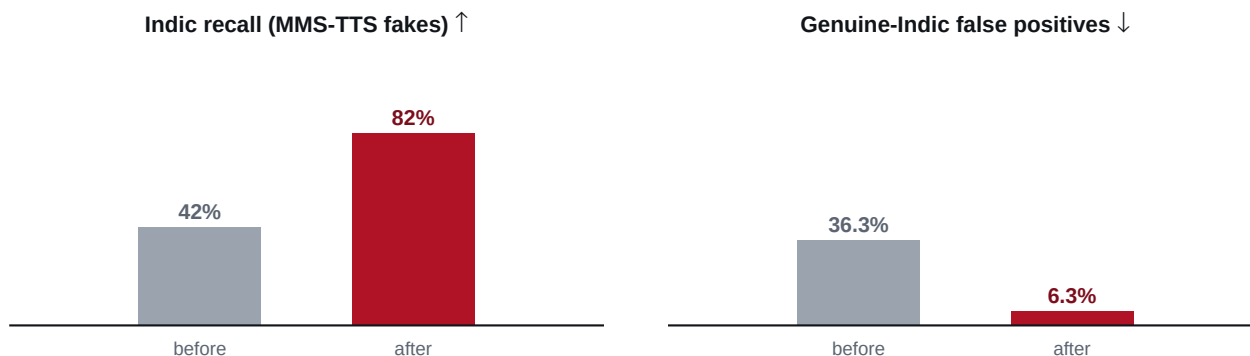


Figure 3 — Indic-aware retraining: recall on MMS-TTS Indic fakes and genuine-Indic false-positive rate.

What it demonstrates: language-specific adaptation can materially change detector behaviour under these conditions, and English EER did not regress (5.9%). **What it does not demonstrate:** it does not establish robustness across all Indic languages, all speakers, other generators, or real deployment — the evidence is specific to MMS-TTS and the tested set.

6 Evaluation Methodology (with honest gaps)

The most useful thing we can show you is exactly how the numbers were obtained — including what we have not pinned down. 'Requires confirmation' marks details not yet documented on our side.

Experiment	Source / generator	Composition	Channel	Train/test split	Metric → result
English detection	In-the-Wild [7]	genuine + spoofed (n: requires confirmation)	clean + augmented	cross-dataset held-out; speaker split: requires confirmation	EER → 5.9%
Detector/fusion	common scored set	per-detector scores	as above	same eval set	EER → 16.0–62.4% / 19.1% / 5.9%
Indic fake recall	MMS-TTS [9]	synthetic Indic fakes	clean	before/after retrain; generator = MMS-TTS only	Recall → 42%→82%
Genuine-Indic FP	genuine Indic speech	10 languages	channel-aware	tested conditions	FPR → 36.3%→6.3%
Calibration	held-out	—	—	—	ECE 0.044 · Brier 0.048

7 Research Questions for Expert Review

These are the questions where your expertise would most change what we do next. We have **not** performed the proposed experiments below; they are how we currently think the claims should be tested.

Q1 — Cross-generator generalisation.

Our Indic evidence uses a single generator (MMS-TTS). What is the most rigorous way to show the detector generalises to unseen synthesis systems? We are considering: train on generators A+B+C, test on held-out D+E, reporting EER, ROC-AUC, TPR@fixed-FPR, and per-generator EER. Is this the right framing, and which generator families would you prioritise?

Q2 — Telephony / codec robustness.

What evaluation protocol would you recommend for 8 kHz G.711, noise and packet loss — clean→codec, a codec→noise→packet-loss cascade, or real captured PSTN/SIP audio? What baselines should accompany it?

Q3 — Generator × codec interaction.

Could a model learn *codec* artifacts rather than *synthesis* artifacts? We propose evaluating a matrix (below) to separate the two. (Proposed, not yet run.)

	Clean	G.711	Noise	Packet loss
Generator A	train	eval	eval	eval
Generator B	train	eval	eval	eval
Generator C	train	eval	eval	eval
Unseen D / E	eval*	eval*	eval*	eval*

*held-out generators — the key generalisation cells. Proposed evaluation design, not results.

Q4 — Speaker disjointness.

We have not yet documented speaker overlap between train and test. Should speaker-disjoint *and* generator-disjoint evaluation be the principal benchmark, to separate spoof detection from speaker memorisation?

Q5 — Disentangling generalisation axes.

How should evaluation separate language vs generator vs speaker vs channel generalisation — is a factorial design practical here?

Q6 — Metric selection for fraud screening.

For real fraud screening, which metrics are most informative — EER, ROC-AUC, PR-AUC, TPR@1%/0.1% FPR, FNR, calibration error, or t-DCF [8]? We currently lead with EER + AUC + ECE and would value your view.

8 Limitations & Failure Modes

Risk	Why it matters	Current evidence	Needed experiment
Unseen generators	attackers use new TTS	MMS-TTS only	generator-disjoint eval
Real-network telephony	augmentation ≠ PSTN	augmented only	captured-call eval
Speaker overlap	inflates EER	not documented	speaker-disjoint split
Very short / noisy speech	calls are messy	not isolated	duration/noise sweep
Code-switching / lang. imbalance	Indic reality	tested subset	per-language study
Adversarial adaptation	attacker adapts	none	adversarial eval

9 Experiment Roadmap (priority-ordered)

#	Experiment	Metric	Interpretation sought
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1	Speaker-disjoint evaluation	EER, AUC	spoof detection vs speaker memorisation
2	Generator-disjoint evaluation	EER by generator	true cross-generator generalisation
3	Codec / channel robustness	EER under G.711/noise/loss	channel vs synthesis cues
4	Per-language Indic evaluation	EER, FPR per language	language-wise reliability
5	Detector-family ablation	Δ EER per family	evidence for fusion hypothesis
6	Calibration analysis	ECE, Brier, reliability curve	probability trustworthiness
7	Real telephone recordings	EER on captured calls	benchmark-to-field gap
8	Adversarial robustness	EER under attack	worst-case behaviour

We deliberately give no expected numbers — these are the experiments, not predictions.

10 Why We Are Writing to You

Your published work on anti-spoofing countermeasures for automatic speaker verification, novel spoof-detection features (cochlear-filter and instantaneous-frequency cues, and group-delay methods), replay-attack and voice-liveness detection, and cross-attack generalisation across the ASVspoof series is directly relevant to the specific questions above. Your feature work on instantaneous-frequency and cochlear-filter cues (CFCCIF) and group-delay methods [15,16] speaks to the phase/HF components of our 89-dim bank; your ASVspoof-generalisation survey [17] frames our central cross-generator question (Q1); your replay / energy-separation instantaneous-frequency work [18] relates to our channel-robustness questions (Q2–Q3); and your voice-liveness (pop-noise) line [19] bears directly on our liveness option and how we should validate it. We are not asking for generic mentorship — we are asking whether our evaluation protocol is scientifically sound and what we should test first.

Representative work we drew on (verified):

- Instantaneous-frequency / cochlear-filter spoof features (CFCCIF), Interspeech 2015 & IEEE SPL 2017 [15,16] — directly relevant to our 89-dim phase / group-delay and HF cues.
- ASVspoof-generalisation survey, APSIPA 2020 [17] — bears on our central cross-generator question (Q1).
- Energy-separation instantaneous-frequency features for replay detection, Interspeech 2019 [18] — relevant to channel/replay robustness (Q2, Q3).
- Pop-noise voice-liveness detection, Springer 2022 [19] — directly relevant to our liveness option and its validation.

What we would value from you (low-friction).

1. Is our current evaluation methodology scientifically sound? 2. What would you change to establish cross-generator generalisation? 3. What telephony/codec evaluation protocol would you recommend? 4. Are we missing important anti-spoofing metrics or baselines? 5. Which failure mode should we investigate first?

Thank you for taking the time to review this work.

Team DigiSeva · VoxShield · GBT BuildStorm 2026

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